

81. Let  $\alpha = \frac{-1+i\sqrt{3}}{2}$ . If  $a = (1+\alpha)\sum_{k=0}^{100}\alpha^{2k}$  and  $b = \sum_{k=0}^{100}\alpha^{3k}$ , then a and b are the roots of the quadratic equation:

माना  $\alpha = \frac{-1+i\sqrt{3}}{2}$  है। यदि  $a = (1+\alpha)\sum_{k=0}^{100}\alpha^{2k}$  तथा  $b = \sum_{k=0}^{100}\alpha^{3k}$ , तो a तथा b निम्न में से किस द्विघात समीकरण के मूल हैं?

(A)  $x^2 + 102x + 101 = 0$

(B)  $x^2 - 101x + 100 = 0$

(C)  $x^2 - 102x + 101 = 0$

(D)  $x^2 + 101x + 100 = 0$

Ans. C

**Sol.**  $\alpha = \frac{-1+i\sqrt{3}}{2} = \omega$

$a = (1 + \omega)(1 + \omega^2 + \omega^4 + \dots + \omega^{200})$

$a = (1 + \omega) \frac{(1 - (\omega^2)^{101})}{(1 - \omega^2)}$

$a = \frac{1 - \omega^{202}}{1 - \omega} = \frac{1 - \omega}{1 - \omega} = 1$

$b = 1 + \omega^3 + \omega^6 + \dots + \omega^{300} = 101$

$S = a + b = 102$

$P = a \cdot b = 101$

Equation.  $x^2 - 102x + 101 = 0$

82. If the set  $\left\{ \operatorname{Re}\left(\frac{z - \bar{z} + z\bar{z}}{2 - 3z + 5\bar{z}}\right) : z \in \mathbb{C}, \operatorname{Re}(z) = 3 \right\}$  is equal to the interval  $(\alpha, \beta]$ , then  $24(\beta - \alpha)$  is equal to :

यदि समुच्चय  $\left\{ \operatorname{Re}\left(\frac{z - \bar{z} + z\bar{z}}{2 - 3z + 5\bar{z}}\right) : z \in \mathbb{C}, \operatorname{Re}(z) = 3 \right\}$  अन्तराल  $(\alpha, \beta]$  के बराबर है, तो  $24(\beta - \alpha)$  का मान होगा :

(A) 27

(B) 42

(C) 36

(D) 30

Ans. D

**Sol.** Let  $z_1 = \left(\frac{z - \bar{z} + z\bar{z}}{2 - 3z + 5\bar{z}}\right)$

Let  $z = 3 + iy$

$\bar{z} = 3 - iy$

$$z_1 = \frac{2iy + (9 + y^2)}{2 - 3(3 + iy) + 5(3 - iy)}$$

$$= \frac{9 + y^2 + i(2y)}{8 - 8iy}$$

$$= \frac{(9 + y^2) + i(2y)}{8(1 - iy)}$$

$$\operatorname{Re}(z_1) = \frac{(9 + y^2) - 2y^2}{8(1 + y^2)}$$

$$= \frac{9 - y^2}{8(1 + y^2)}$$

$$= \frac{1}{8} \left[ \frac{10 - (1 + y^2)}{(1 + y^2)} \right]$$

$$= \frac{1}{8} \left[ \frac{10}{1 + y^2} - 1 \right]$$

$$1 + y^2 \in [1, \infty]$$

$$\frac{1}{1 + y^2} \in (0, 1]$$

$$\frac{10}{1 + y^2} \in (0, 10]$$

$$\frac{10}{1 + y^2} - 1 \in (-1, 9]$$

$$\operatorname{Re}(z_1) \in \left( \frac{-1}{8}, \frac{9}{8} \right]$$

$$\alpha = \frac{-1}{8}, \beta = \frac{9}{8}$$

$$24(\beta - \alpha) = 24 \left( \frac{9}{8} + \frac{1}{8} \right) = 30$$

83. Let  $9 = x_1 < x_2 < \dots < x_7$  be in an A.P. with common difference  $d$ . If the standard deviation of  $x_1, x_2, \dots, x_7$  is 4 and the mean is  $\bar{x}$ , then  $\bar{x} + x_6$  is equal to :

माना  $9 = x_1 < x_2 < \dots < x_7$  समान्तर श्रेणी में हैं, जिसका सार्वान्तर  $d$  है। यदि  $x_1, x_2, \dots, x_7$  का मानक विचलन 4 तथा माध्य  $\bar{x}$  तब  $\bar{x} + x_6$  बराबर होगा :

- (A) 25                      (B)  $2\left(9 + \frac{8}{\sqrt{7}}\right)$                       (C)  $18\left(1 + \frac{1}{\sqrt{3}}\right)$                       (D) 34

**Ans.** D

**Sol.** Mean  $\Rightarrow \bar{x} = \frac{\sum_{i=1}^7 x_i}{7} = \frac{7[2a + 6d]}{7} = a + 3d = x_4$

$$\text{Variance} = \frac{\sum_{i=1}^7 (x_i - \bar{x})^2}{7} = (4)^2 \Rightarrow \frac{\sum_{i=1}^7 (x_i - x_4)^2}{7} = 16$$

$$\Rightarrow \frac{(3d)^2 + (2d)^2 + d^2 + 0 + d^2 + (2d)^2 + (3d)^2}{7} = 16$$

$$= 4d^2 = 16 \quad \Rightarrow \quad d = 2$$

$$\Rightarrow \bar{x} = 9 + 3(2) = 15$$

$$\& x_6 = a + 5d = 9 + 5(2) = 19 \Rightarrow \bar{x} + x_6 = 34$$

84. The total number of positive integral solutions  $(x, y, z)$  such that  $xyz = 24$  is :

$xyz = 24$  के धन पूर्णांक हलों  $(x, y, z)$  की कुल संख्या है :

- (A) 24                      (B) 36                      (C) 45                      (D) 30

**Ans.** D

**Sol.**  $xyz = 24 = 2^3 \times 3$

Power of 2 & 3 are to be distributed among  $x, y, z$

Powers of 2 :

$$\text{Number of ways} = {}^5C_2 = 10$$

Powers of 3 :

$$\text{Number of ways} = {}^3C_2 = 3$$

$$\text{Required number of ways} = 10 \times 3 = 30$$

85. The probability that a randomly chosen 5-digit number is made from exactly two digits is :

यादृच्छया चुनी गई पाँच अंकों की एक संख्या के मात्र दो अंको से बनाई गई होने की प्रायिकता है :

- (A)  $\frac{150}{10^4}$  (B)  $\frac{121}{10^4}$  (C)  $\frac{134}{10^4}$  (D)  $\frac{135}{10^4}$

Ans. D

Sol. 

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

$$9 \times 10 \times 10 \times 10 \times 10 = 9 \times 10^4$$

Required way

Without '0' (1, 2, ..., 9)

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

$${}^9C_2 (2^5 - 2) = 1080$$

with '0' (1, 2, ..., 9)

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

$$= {}^9C_1 (1 \times 2^4 - 1) = 135$$

$$\text{Ans} = \frac{1080 + 135}{9 \times 10^4}$$

$$= \frac{135}{10^4}$$

86. Let A denote the event that a 6 digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, be divisible by 3. Then probability of event A is equal to:

माना A, अंकों 0, 1, 2, 3, 4, 5, 6 द्वारा बिना पुनरावृत्ति के बनाई गई 6 अंकों की संख्या के 3 से विभाजित होने की घटना को दर्शाता है। तो घटना A की प्रायिकता बराबर है :

- (A)  $\frac{11}{27}$  (B)  $\frac{9}{56}$  (C)  $\frac{3}{7}$  (D)  $\frac{4}{9}$

Ans. D

Sol. Case I : 1, 2, 3, 4, 5, 6

$$\text{Total} = 6!$$

Case II : 0, 1, 2, 4, 5, 6

$$\text{Total} = 5 \times 5!$$

Case III : 0, 1, 2, 3, 4, 5

$$\frac{\text{Total} = 5 \times 5!}{16 \times 5!}$$

$$P(A) = \frac{16 \times 5!}{6 \times (6 \times 5 \times 4 \times 3 \times 2)} = \frac{4}{9}$$

87. If an unbiased die, marked with -2, -1, 0, 1, 2, 3 on its faces, is thrown five times, then the probability that the product of the outcomes is positive, is :

यदि एक अनभिन्नत पासे, जिसके फलकों पर -2, -1, 0, 1, 2, 3 लिखा है, को पाँच बार फेंका जाता है, तो फलकों पर प्राप्त संख्याओं का गुणनफल धनात्मक होने की प्रायिकता है :

- (A)  $\frac{440}{2592}$  (B)  $\frac{27}{288}$  (C)  $\frac{521}{2592}$  (D)  $\frac{881}{2592}$

**Ans.** C

**Sol.** Either all outcomes are positive or any two are negative.

$$\text{Now, } p = P(\text{positive}) = \frac{3}{6} = \frac{1}{2}$$

$$q = p(\text{negative}) = \frac{2}{6} = \frac{1}{3}$$

Required probability

$$= {}^5C_5 \left(\frac{1}{2}\right)^5 + {}^5C_2 \left(\frac{1}{3}\right)^2 \left(\frac{1}{2}\right)^3 + {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{1}{2}\right)^1$$

$$= \frac{521}{2592}$$

88. A data consists of  $n$  observations :  $x_1, x_2, \dots, x_n$ . If  $\sum_{i=1}^n (x_i + 1)^2 = 9n$  and  $\sum_{i=1}^n (x_i - 1)^2 = 5n$ , then standard deviation of this data is :

आंकड़ों के एक समूह में  $n$  प्रेक्षण  $x_1, x_2, \dots, x_n$  है। यदि  $\sum_{i=1}^n (x_i + 1)^2 = 9n$  तथा  $\sum_{i=1}^n (x_i - 1)^2 = 5n$  है तो इन आंकड़ों का मानक विचलन है —

- (A)  $\sqrt{5}$  (B) 5 (C)  $\sqrt{7}$  (D) 2

**Ans.** A

Sol. 
$$\left. \begin{aligned} \sum (x_i + 1)^2 &= 9n = \sum x_i^2 + 2 \sum x_i + \sum 1 \\ \sum (x_i - 1)^2 &= 5n = \sum x_i^2 - 2 \sum x_i + \sum 1 \end{aligned} \right\} \Rightarrow \sum x_i^2 = 6n$$

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = \sqrt{\frac{\sum x_i^2 - 2(\sum x_i)\bar{x} + \sum \bar{x}}{n}} = \sqrt{\frac{6n - 2n + n}{n}} = \sqrt{5}$$

$$\bar{x} = \frac{\sum x_i}{n} = 1$$

89. The value of  $(2.^1P_0 - 3.^2P_1 + 4.^3P_2 - \dots \text{ up to } 51^{\text{th}} \text{ term}) + (1! - 2! + 3! - \dots \text{ up to } 51^{\text{th}} \text{ term})$  is equal to:  
 $(2.^1P_0 - 3.^2P_1 + 4.^3P_2 - \dots 51 \text{ वें पद तक}) + (1! - 2! + 3! - \dots 51 \text{ वें पद तक})$  का मान बराबर:  
 (A)  $1 - 51(51)!$  (B)  $1 + (52)!$  (C) 1 (D)  $1 + (51)!$

Ans. B

Sol.  $(2.1 - 3.2! + 4.3! - 5.4! + \dots 51 \text{ terms})$   
 $+ (1! - 2! + 3! - 4! + \dots 51 \text{ terms})$   
 $\Rightarrow (2! - 3! + 4! - 5! + \dots + 52!)$   
 $+ (1! - 2! + 3! - 4! + \dots + 51!)$   
 $\Rightarrow 1! + 52! = 1 + 52!$

### INTEGER TYPE QUESTIONS

90. Let  $z = 1 + i$  and  $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$ . Then  $\frac{12}{\pi} \arg(z_1)$  is equal to \_\_\_\_\_.

माना  $z = 1 + i$  तथा  $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$  है तो  $\frac{12}{\pi} \arg(z_1)$  बराबर है।

Ans. 9

Sol.  $z = 1 + i$

$$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$$

$$z_1 = \frac{1+i(1-i)}{(1-i)(1-1-i) + \frac{1}{1+I}}$$

$$= \frac{1+i-i^2}{(1-i)(-i) + \frac{1-i}{2}}$$

$$= \frac{2+i}{\frac{-3i-1}{2}} = \frac{4+2i}{-3i-1}$$

$$= \frac{-(4+2i)(3i-1)}{(3i)^2 - (1)^2}$$

$$\text{Arg}(z_1) = \frac{3\pi}{4}$$

$$\therefore \frac{12}{\pi} \arg(z_1) = \frac{12}{\pi} \times \frac{3\pi}{4} = 9$$

91. The number of times the digit 3 will be written when listing the integers from 1 to 1000 is :

1 से 1000 तक के पूर्णाकों को क्रम से लिखने पर अंक 3, \_\_\_\_\_ बार लिखा जायेगा।

Ans. 300

Sol. Number of times; digit 3 will be written =  $10 \times 10 + 10 \times 10 + 10 \times 10$

$$\frac{3}{10} \times \frac{10}{10} + \frac{3}{10} \times \frac{10}{10} + \frac{3}{10} \times \frac{10}{10}$$

92. If  $\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{i-1}\right)^{n/3} = 1, (m, n \in \mathbf{N})$ , then the greatest common divisor of the least values of m and n is \_\_\_\_\_.

यदि  $\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{i-1}\right)^{n/3} = 1, (m, n \in \mathbf{N})$  तो m तथा n के न्यूनतम मानों का महत्तम उभयनिष्ठ भाजक है \_\_\_\_\_।

Ans. 4

Sol.  $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{-1+i}\right)^{\frac{n}{3}} = 1$

$$(i)^{\frac{m}{2}} = (-i)^{\frac{n}{3}} = 1$$

Least value  $m = 8$

$$n = 12$$

$$\text{HCF}(8, 12) = 4$$

93. The sum of the series  $2 \times 1 \times {}^{20}C_4 - 3 \times 2 \times {}^{20}C_5 + 4 \times 3 \times {}^{20}C_6 - 5 \times 4 \times {}^{20}C_7 + \dots + 18 \times 17 \times {}^{20}C_{20}$  is equal to \_\_\_\_\_.

श्रेणी  $2 \times 1 \times {}^{20}C_4 - 3 \times 2 \times {}^{20}C_5 + 4 \times 3 \times {}^{20}C_6 - 5 \times 4 \times {}^{20}C_7 + \dots + 18 \times 17 \times {}^{20}C_{20}$  का योग है \_\_\_\_\_.

**Ans.** 34

**Sol.** Let  $p = \sum_{r=1}^{17} (-1)^{r+1} \cdot r \cdot (r+1) \cdot {}^{20}C_{r+3}$

$$= \sum_{r=1}^{17} (-1)^{r+1} ((r+3)-3)((r+3)-2) {}^{20}C_{r+3}$$

$$p = \sum_{r=1}^{17} (-1)^{r+1} ((r+3)^2 - 5(r+3) + 6) {}^{20}C_{r+3} \quad \dots(1)$$

Now,

$$(r+3)^2 \cdot {}^{20}C_{r+3} = (r+3) \cdot 20 \cdot {}^{19}C_{r+2} = 20((r+2)+1) {}^{19}C_{r+2}$$

$$= 380 \cdot {}^{18}C_{r+1} + 20 \cdot {}^{19}C_{r+2} \quad \dots(2)$$

and

$$(r+3) \cdot {}^{20}C_{r+3} = 20 \cdot {}^{19}C_{r+2} \quad \dots(3)$$

from (1), (2) and (3)

$$p = \sum_{r=1}^{17} (-1)^{r+1} [380 \cdot {}^{18}C_{r+1} - 80 \cdot {}^{19}C_{r+2} + 6 \cdot {}^{20}C_{r+3}]$$

$$= 380({}^{18}C_2 - {}^{18}C_3 + \dots + {}^{18}C_{18})$$

$$- 80({}^{19}C_3 - {}^{19}C_4 + \dots + {}^{19}C_{19})$$

$$+ 6({}^{20}C_4 - {}^{20}C_5 + \dots + {}^{20}C_{20})$$

$$= 380({}^{18}C_1 - {}^{18}C_0) - 80({}^{19}C_0 - {}^{19}C_1 + {}^{19}C_2) + 6({}^{20}C_0 + {}^{20}C_1 - {}^{20}C_2 + {}^{20}C_3)$$

$$= 380(17) - 80(153) + 6(969)$$

$$= 34$$

94. Let  $a = 1 + \frac{{}^2C_2}{3!} + \frac{{}^3C_2}{4!} + \frac{{}^4C_2}{5!} + \dots$ ,

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \dots$$

Then  $\frac{2b}{a^2}$  is equal to \_\_\_\_\_.

माना  $a = 1 + \frac{{}^2C_2}{3!} + \frac{{}^3C_2}{4!} + \frac{{}^4C_2}{5!} + \dots$ ,

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \dots$$

हैं, तो  $\frac{2b}{a^2}$  बराबर है \_\_\_\_\_

**Ans.** 8

**Sol.**  $a = 1 + \frac{{}^2C_2}{3} + \frac{{}^3C_2}{4} + \frac{{}^4C_2}{5} + \dots$

$$a = 1 + \sum_{r=2}^{\infty} \frac{{}^rC_2}{(r+1)}$$

$$a = 1 + \sum_{r=2}^{\infty} \frac{r(r-1)}{2(r+1)}$$

$$a = 1 + \frac{1}{2} \sum_{r=2}^{\infty} \frac{r(r+1) - 2r}{(r+1)}$$

$$a = 1 + \frac{1}{2} \left[ \sum_{r=2}^{\infty} \frac{1}{(r-1)} - \frac{2}{r} + \frac{2}{(r+1)} \right]$$

$$a = 1 + \frac{1}{2} \left[ (e-1) - 2(e-2) + 2 \left( e - 2 - \frac{1}{2} \right) \right]$$

$$a = 1 + \frac{1}{2} [e - 2] = \frac{e}{2}$$

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \dots$$

$$b = e^2$$

$$\text{Now, } = \frac{2b}{a^2} = \frac{2 \times e^2}{e^2} \times 4 = 8$$

95. Let  $x$  and  $y$  be distinct integers where  $1 \leq x \leq 25$  and  $1 \leq y \leq 25$ . Then, the number of ways of choosing  $x$  and  $y$ , such that  $x + y$  is divisible by 5, is \_\_\_\_\_.

माना  $x$  तथा  $y$  भिन्न पूर्णांक हैं जहाँ  $1 \leq x \leq 25$  तथा  $1 \leq y \leq 25$ । तब  $x$  तथा  $y$  के चुने जाने के तरीकों की संख्या, जबकि  $x + y$ , 5 से विभाज्य हो।

**Ans.** 120

**Sol.** we divides numbers  $1 \leq x \leq 25$ ,  $1 \leq y \leq 25$ ,  $x \in \mathbb{N}$  in to following five groups

| Ist | II <sup>nd</sup> | III <sup>rd</sup> | IV <sup>th</sup> | V <sup>th</sup> |
|-----|------------------|-------------------|------------------|-----------------|
| 1   | 2                | 3                 | 4                | 5               |
| 6   | 7                | 8                 | 9                | 10              |
| 11  | 12               | 13                | 14               | 15              |
| 16  | 17               | 18                | 19               | 20              |
| 21  | 22               | 23                | 24               | 25              |

$x + y$  is divisible by 5 if

$$x \in \text{I}^{\text{st}} \text{ and } y \in \text{IV}^{\text{th}} \Rightarrow {}^5C_1 \times {}^5C_1 = 25$$

$$x \in \text{II}^{\text{nd}} \text{ and } y \in \text{III}^{\text{rd}} \Rightarrow {}^5C_1 \times {}^5C_1 = 25$$

$$x \in \text{III}^{\text{rd}} \text{ and } y \in \text{II}^{\text{nd}} \Rightarrow {}^5C_1 \times {}^5C_1 = 25$$

$$x \in \text{IV}^{\text{th}} \text{ and } y \in \text{I}^{\text{st}} \Rightarrow {}^5C_1 \times {}^5C_1 = 25$$

$$x \in \text{V}^{\text{th}} \text{ and } y \in \text{V}^{\text{th}} \Rightarrow {}^5C_2 \times 2 = 20$$

Total = 120

96. Let  $(1 + x + x^2)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20}$ . If  $(a_1 + a_3 + a_5 + \dots + a_{19}) - 11a_2 = 121k$ , then  $k$  is equal to \_\_\_\_\_.

माना  $(1 + x + x^2)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20}$  है। यदि  $(a_1 + a_3 + a_5 + \dots + a_{19}) - 11a_2 = 121k$  है, तो  $k$  बराबर है \_\_\_\_\_.

**Ans.** 239

**Sol.**  $x = 1$   $a_0 + a_1 + a_2 + \dots + a_{19} + a_{20} = 3^{10}$

$x = -1$   $a_0 - a_1 + a_2 - \dots + a_{19} + a_{20} = 1$

$$\Rightarrow 2(a_1 + a_3 + \dots + a_{19}) = 3^{10} - 1$$

diff.  $10(1 + x + x^2)^9 (1 + 2x) = a_1 + 2a_2x + 3a_3x^2 + \dots$

$$\text{diff. } 90(1+x+x^2)^8(1+2x)^2 + 10(1+x+x^2)^9 \cdot 2 = 2a_2 + 6a_3x + \dots$$

$$\text{Put } x = 0 \Rightarrow 90 + 20 = 2a_2 \Rightarrow a_2 = 55$$

$$\text{So, } a_1 + a_3 + \dots + a_{19} - 11a_2 = \frac{3^{10}-1}{2} - 11 \times 55 = 121k$$

$$k = 239$$

97. In the expansion of  $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$ ,  $x \neq 0$ , the sum of the coefficients of  $x^3$  and  $x^{-13}$  is equal to \_\_\_\_\_.

$$(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5, x \neq 0, \text{ के प्रसार में } x^3 \text{ तथा } x^{-13} \text{ के गुणांकों का योग है } \underline{\hspace{2cm}}$$

**Ans.** 118

$$\text{Sol. } (1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$$

$$= (1+x)(1-x^2)\left(\left(1+\frac{1}{x}\right)^3\right)^5$$

$$= \frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}}$$

$$= \frac{(1+x)^{17} - x(1+x)^{17}}{x^{15}}$$

$$= \text{coeff}(x^3) \text{ in the expansion} = \text{coeff}(x^{18}) \text{ in } (1+x)^{17} - x(1+x)^{17}$$

$$= 0 - 1$$

$$= -1$$

$$\text{coeff}(x^{-13}) \text{ in the expansion} = \text{coeff}(x^2) \text{ in } (1+x)^{17} - x(1+x)^{17}$$

$$= \binom{17}{2} - \binom{17}{1}$$

$$= 17 \times 8 - 17$$

$$= 17 \times 7$$

$$= 119$$

Hence Answer =  $119 - 1 = 118$

98. If all the six digit numbers  $x_1 x_2 x_3 x_4 x_5 x_6$  with  $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$  are arranged in the increasing order, then the sum of the digits in the 72<sup>th</sup> number is \_\_\_\_\_ .

यदि सभी छः अंकों की संख्याओं  $x_1 x_2 x_3 x_4 x_5 x_6$ , जिनमें  $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$  है, को वर्धमान क्रम में व्यवस्थित किया जाता है, तब 72<sup>वीं</sup> संख्या में अंकों का योग है।

**Ans.** 32

**Sol.** Number of numbers starting with 1 =  ${}^8C_5 = 56$

Number of numbers starting with 23 =  ${}^6C_4 = 15$

Next number at 72<sup>nd</sup> option is 245678.

$\therefore$  sum of digits =  $2 + 4 + 5 + 6 + 8 = 32$

99. Let the coefficient of  $x^r$  in the expansion of

$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$  be  $\alpha_r$ . If  $\sum_{r=0}^n \alpha_r = \beta^n - \gamma^n$ ,  $\beta, \gamma \in \mathbb{N}$ , then the value of  $\beta^2 + \gamma^2$  equals \_\_\_\_\_.

माना  $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$  के प्रसार में  $x^r$  का गुणांक  $\alpha_r$  है। यदि

$\sum_{r=0}^n \alpha_r = \beta^n - \gamma^n$ ,  $\beta, \gamma \in \mathbb{N}$  है, तो  $\beta^2 + \gamma^2$  बराबर है \_\_\_\_\_

**Ans.** 25

**Sol.**  $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$

$$= (x+3)^{n-1} \frac{\left\{ 1 - \left( \frac{x+2}{x+3} \right)^n \right\}}{1 - \left( \frac{x+2}{x+3} \right)}$$

$$= (x+3)^n - (x+2)^n$$

$$\sum_{r=0}^n \alpha_r = \sum_{r=0}^n [(x+3)^n - (x+2)^n] = \beta^n - \gamma^n$$

$$\Rightarrow 4^n - 3^n = \beta^n - \gamma^n$$

$$\Rightarrow \beta = 4, \gamma = 3$$

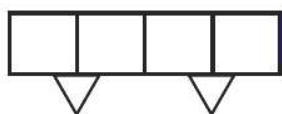
$$\beta^2 + \gamma^2 = 25$$

100. A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is \_\_\_\_\_.

एक व्यक्ति अपने चार अंक का ATM पिन कोड भूल जाता है, लेकिन उसे यह याद रहता है कि सभी अंक भिन्न-भिन्न हैं, सबसे बड़ा अंक 7 है तथा प्रथम दो अंकों का योग अन्तिम दो अंकों के योग के बराबर है, तो सही कोड के लिए आवश्यक अधिकतम प्रयासों की संख्या होगी \_\_\_\_\_

**Ans.** 72

**Sol.**



Sum of first two digits = Sum of last two digits =  $\alpha$

C-I :  $\alpha = 7$

$2 \times 12 = 24$  ways

|   |   |   |   |
|---|---|---|---|
| 7 | 0 |   |   |
| 0 | 7 | 1 | 6 |
|   |   | 2 | 5 |
|   |   | 3 | 4 |
|   |   | 4 | 3 |
|   |   | 5 | 2 |
|   |   | 6 | 1 |

C-II :  $\alpha = 8$



17

71



26

62

35

53



$2 \times 8$  Ways

= 16 ways

Case-III :  $\alpha = 9$

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| 2 | 7 | 3 | 6 |
| 7 | 2 | 6 | 3 |
|   |   | 4 | 5 |
|   |   | 5 | 4 |

$\square \square \square \square$

$2 \times 8$  Ways

= 16 ways

Case-IV :  $\alpha = 10$

$\square \square \quad \square \square$

37      46

73      64

$2 \times 4$  ways

8 ways

Case-V :  $\alpha = 11$

$\square \square \quad \square \square$

47      56

74      65

$2 \times 4$  ways

= 8 ways

Ans.  $24 + 16 + 16 + 8 + 8 = 72$